

WAVE TRANSMISSION BY FLOATING BREAKWATERS USING SPH-BASED NUMERICAL ANALYSIS

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OVERVIEW

Motivation

Floating breakwaters are widely used for coastal protection, yet design remains based on empirical transmission curves that do not capture nonlinear wave–structure interaction or mooring dynamics

Objective

To develop a calibrated SPH framework to evaluate how structural constraints and mooring systems influence wave transmission.

Research Questions

- ➔ **MAIN RQ:** To what extent can nonlinear SPH simulations improve the prediction of wave transmission in floating breakwaters compared to classical design relationships based on B/L and C_t ?
- ➔ **SUB –RQ’s**
 - How does restricting rigid-body motion (fixed vs free) influence transmission?
 - How do taut and catenary mooring systems modify wave–structure interaction and transmission behavior?
 - Does the classical relationship between C_t and B/L remain valid under nonlinear, motion-coupled conditions?

Methodology

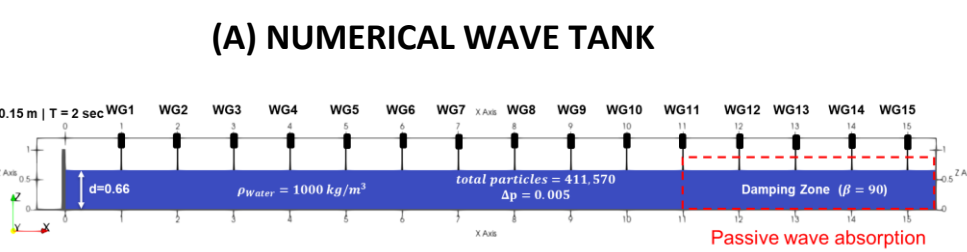


Figure 1. Numerical wave–tank layout used during calibration

Wave Tank Setup

- Water depth: $d = 0.66$ m
- Wave conditions: $H = 0.15$ m, $T = 2.0$ s
- Piston wavemaker with passive damping zone

Model Calibration

- Spatial resolution: $dp = 0.01$ m
- Time stepping: $CFL = 0.20$
- Damping parameter: $\beta = 10$
- Calibration ensured stable wave propagation with minimal reflection

Test Configurations

- Fixed structure
- Free-floating structure
- Taut-moored system
- Catenary-moored system

Wave transmission was evaluated using:

$$C_t = \frac{H_t}{H_i}$$

(Goda and Suzuki, 1976) reflection, incident wave separation used to determine C_t and C_r , which are transmission coefficient and reflection coefficient respectively.

(B) MODEL CALIBRATION

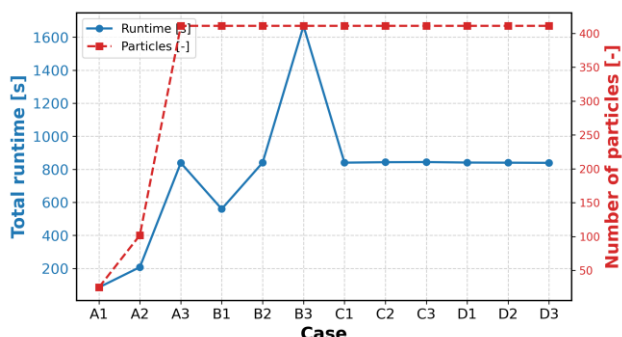


Figure 2. Total runtime and particle count for all calibration families

Table 1: Summary of bare-tank calibration cases, showing particle spacing dp , CFL, damping coefficient (β), total runtime, and particle count

Case	dp [m]	CFL [-]	β [-]	Runtime [s]	Particles [-]
A1	0.02	0.2	10	85.09	25,769
A2	0.01	0.2	10	207.47	102,986
A3	0.005	0.2	10	838.90	411,570
B1	0.005	0.3	10	559.58	411,570
B2	0.005	0.2	10	840.05	411,570
B3	0.005	0.1	10	1668.35	411,570
C1	0.005	0.2	10	840.03	411,570
C2	0.005	0.2	10	842.81	411,570
C3	0.005	0.2	10	843.80	411,570
D1	0.005	0.2	5	840.66	411,570
D2	0.005	0.2	10	839.90	411,570
D3	0.005	0.2	20	838.91	411,570

Spatial resolution: $dp=0.01$ m

Time stepping: $CFL = 0.20$

Damping coefficient: $\beta=10$

Convergence verified using wave gauge measurements

(C) TEST CONFIGURATIONS

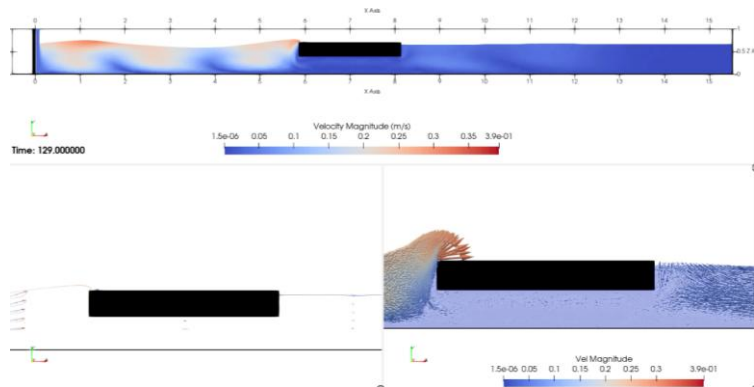


Figure 3 Fixed case

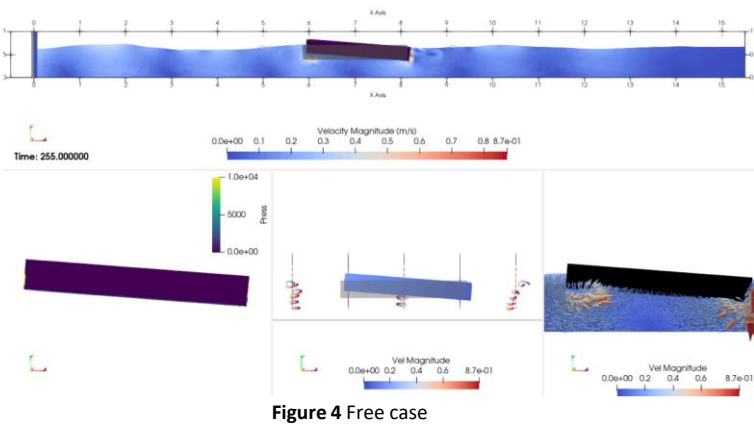


Figure 4 Free case

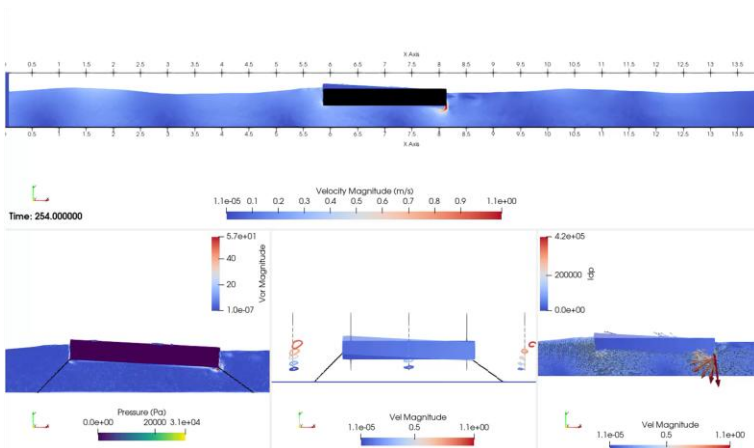


Figure 5 Taut-Moored Case

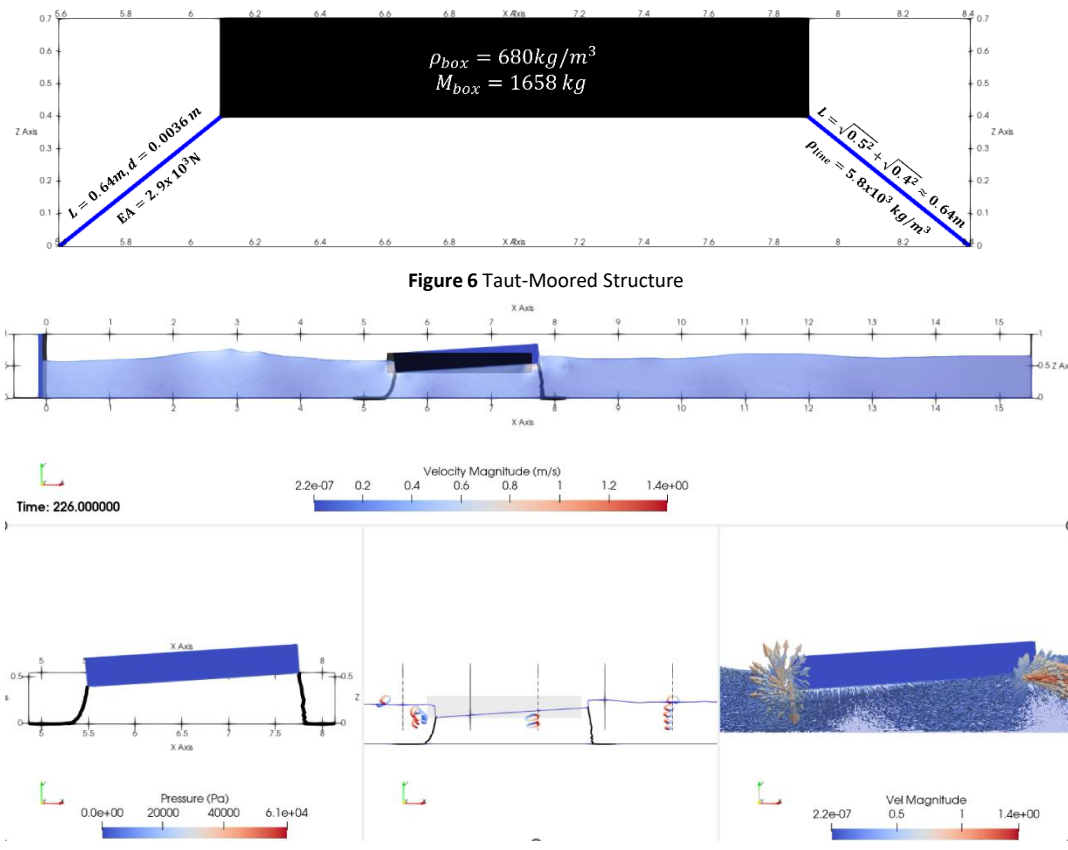


Figure 6 Taut-Moored Structure

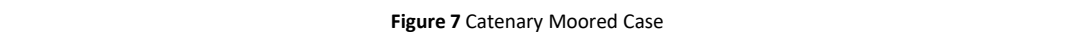


Figure 7 Catenary Moored Case

Results

Fixed Structure

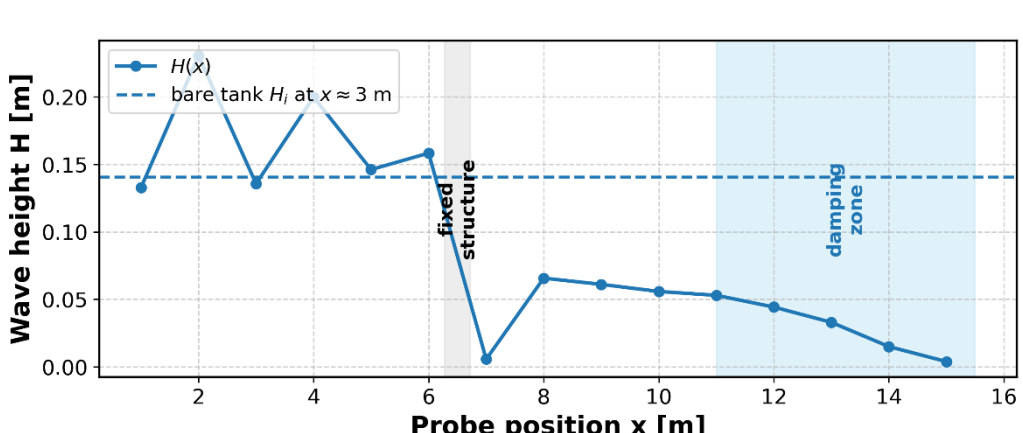


Figure 8 Catenary Moored Case

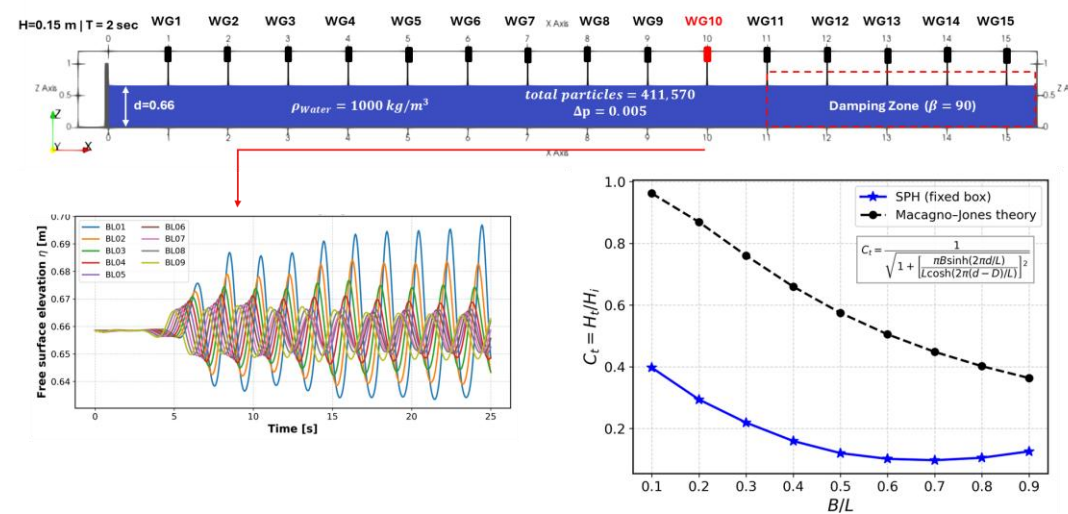
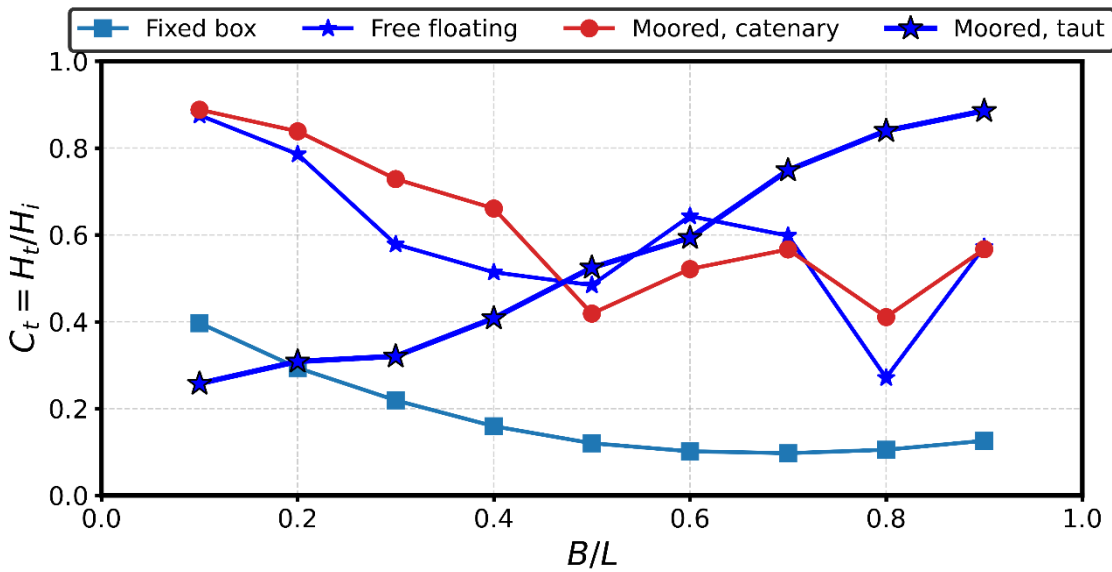
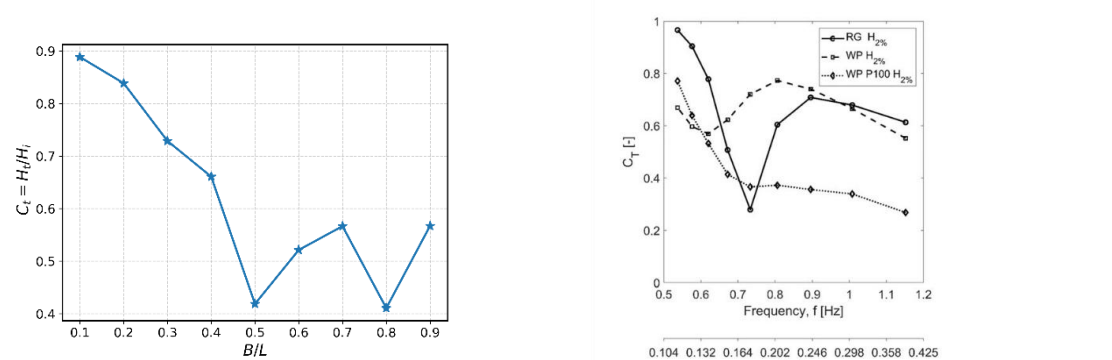


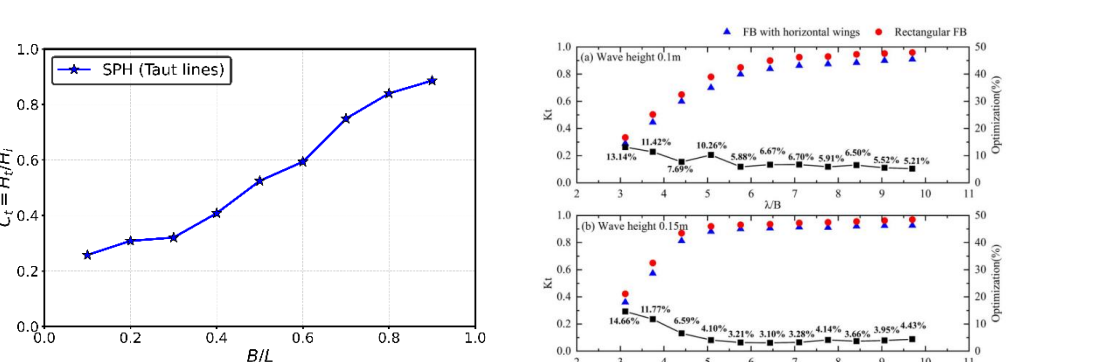
Figure 9 Catenary Moored Case



Discussion and Conclusion

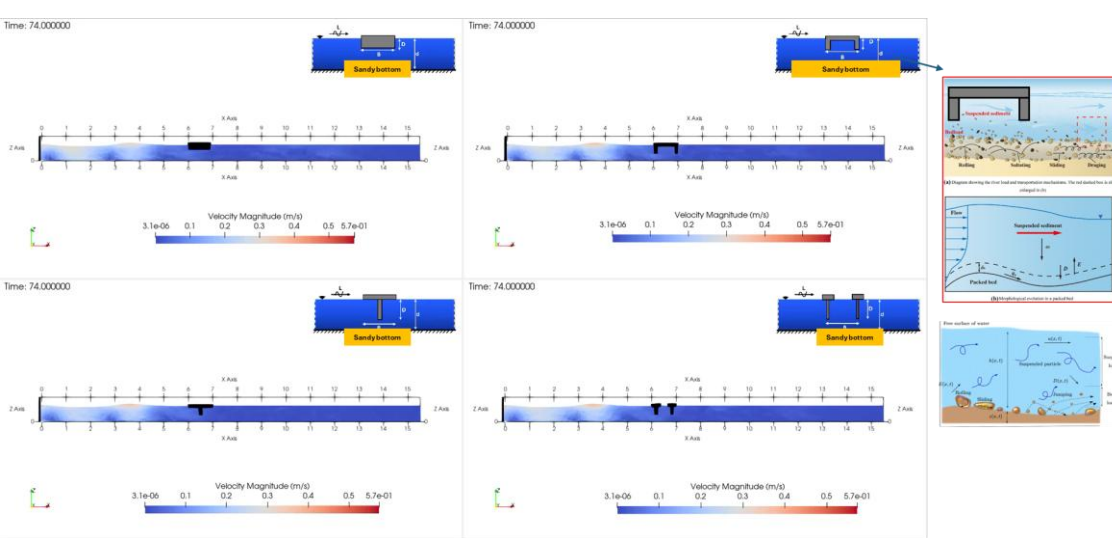


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Future Work



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References

